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Impulse response of continuous LTI systems

※ Definition of impulse response

※ Solution of impulse responses

1. Definition of impulse response

Under the condition that the initial state of the system is zero, the output response generated by excitation of the system δ (t) with the impulse signal is called the impulse response of the system, expressed by the symbol h(t).

(System initial state is zero)

The impulse response h(t) of the system reflects the time domain characteristics of a continuous LTI system. It is only related to the characteristics of the system itself, and has nothing to do with the inputs and outputs of the system.

1. Definition of impulse response

If the linear constant coefficient differential equation is described as a continuous LTI system

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)$$
$$=b_mx^{(m)}(t)+b_{m-1}x^{(m-1)}(t)+\cdots+b_1x'(t)+b_0x(t)$$

Then the impulse response h(t) of the continuous LTI system should be satisfied

$$h^{(n)}(t)+a_{n-1}h^{(n-1)}(t)+\cdots+a_1h'(t)+a_0h(t)$$
$$=b_m\delta^{(m)}(t)+b_{m-1}\delta^{(m-1)}(t)+\cdots+b_1\delta'(t)+b_0\delta(t)$$

When t > 0, the function values of $\delta(t)$ and its derivative are equal to zero.

2. Solving the impulse response

[Example] Describes the differential equation of a continuous LTI system as

$y'(t)+3y(t)=2x(t), \quad t>0$ Try to find the impulse response h(t) of the system.

analyse: When the system input signal x(t) is d(t) and the output signal y(t) is h(t)

The differential equation that describes the system is

$$h'(t)+3h(t)=2\delta(t)$$

When t > 0, $\delta(t)$=0, the differential equation describing the system is

$$h'(t)+3h(t)=0$$

Therefore, the impulse response of the system has the form of homogeneous solution

2. Solving the impulse response

[Example] Describes the differential equation of a continuous LTI system as

$$y'(t)+3y(t)=2x(t), \quad t>0$$ Try to find the impulse response of the system.

How do I determine the pending coefficient of a homogeneous solution?

Substitute homogeneous solutions into differential equations $h'(t)+3h(t)=2\delta(t)$

Balance the terms of the same kind on both sides of the equation, find the coefficient to be determined, and get h(t).

This method is called the
impulse balancing method

[Example] Describes the differential equation of a continuous LTI system as

$y'(t)+3y(t)=2x(t), t>0$ Try to find the impulse response of the system

Solution: $h'(t)+3h(t)=2\delta(t)$

Feature root $s=-3$ $\Rightarrow$ $h(t)=Ae^{-3t}u(t)$

Substitute differential equations $h'(t)+3h(t)=2\delta(t)$

$$[Ae^{-3t}u(t)]'+3Ae^{-3t}u(t)=2\delta(t)$$

$$-3Ae^{-3t}u(t)+A\delta(t)+3Ae^{-3t}u(t)=2\delta(t)$$

$$A\delta(t)=2\delta(t) \Rightarrow A=2$$

$$h(t)=2e^{-3t}u(t)$$

If $y'(t)+3y(t)=2x(t)+x'(t)$, $t>0$

How does the impulse response change?

$$h'(t)+3h(t)=2\delta(t)+\delta'(t)$$

Feature roots = -3, if still set $h(t)=Ae^{-3t}u(t)$

Substitute $h'(t)+3h(t)=2\delta(t)+\delta'(t)$ to get:

$$A\delta(t)=2\delta(t)+\delta'(t)$$

The two ends of the equation cannot be balanced!

If it is balanced, h(t) needs to add a derivative term of u(t), $\delta(t)$

$$h(t)=Ae^{-3t}u(t)+B\delta(t)$$

If $y'(t)+3y(t)=2x(t)+x'(t)\quad,\quad t>0$

How does the impulse response change?

$$h(t)=Ae^{-3t}u(t)+B\delta(t)$$

Substitute $h'(t)+3h(t)=2\delta(t)+\delta'(t)$

then $[Ae^{-3t}u(t)+B\delta(t)]'+3[Ae^{-3t}u(t)+B\delta(t)]=2\delta(t)+\delta'(t)$

$$A\delta(t)+3B\delta(t)+B\delta'(t)=2\delta(t)+\delta'(t)$$

Then $A=-1,\ B=1$

$$h(t)=-e^{-3t}u(t)+\delta(t)$$

The form of the visible impulse response is set according to the situation of the differential equations

2. Solving the impulse response

The impulse response h(t) of a continuous LTI system satisfies the differential equations

$$h^{(n)}(t)+a_{n-1}h^{(n-1)}(t)+\cdots+a_1h'(t)+a_0h(t)$$
$$=b_m\delta^{(m)}(t)+b_{m-1}\delta^{(m-1)}(t)+\cdots+b_1\delta'(t)+b_0\delta(t)$$

(1) when ***n>m*** (It is assumed that the feature root is an unequal solid root)

$$h(t)=(\sum_{i=1}^{n}K_i\mathrm{e}^{s_it})u(t)$$

(2) When **n≤m** h(t) should contain the impulse and its higher derivative

$$h(t)=(\sum_{i=1}^{n}K_i\mathrm{e}^{s_it})u(t)+\sum_{j=0}^{m-n}A_j\delta^{(j)}(t)$$

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$$h(t)=(\sum_{i=1}^{n}K_i\mathrm{e}^{s_it})u(t)+\sum_{j=0}^{m-n}A_j\delta^{(j)}(t)$$

The exponential form before u(t) is determined by the eigen of the differential equation (assuming an unequal solid root).

The term $\delta^{(j)}(t)$ is determined by the highest derivative of the differential equation $\delta(t)$ and the highest derivative of h(t).

2. Solving the impulse response

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Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!